

4

Pipe Analysis

In this chapter we focus our attention mainly on the strength capabilities of a pipeline. We discuss different materials used to construct pipelines and how to calculate the amount of internal pressure that a given pipe can withstand. We determine the amount of internal pressure a particular size of pipe can withstand based on the pipe material, diameter, and wall thickness. Next, we establish the hydrostatic test pressure the pipeline will be subject to, such that the previously calculated internal pressure can be safely tolerated. We also discuss the volume content or line fill volume of a pipeline and how it is used in batched pipelines with multiple products.

4.1 Allowable Operating Pressure and Hydrostatic Test Pressure

To transport a liquid through a pipeline, the liquid must be under sufficient pressure so that the pressure loss due to friction and the pressure required for any elevation changes can be accommodated. The longer the pipeline and the higher the flow rate, the higher the friction drop will be, requiring a corresponding increase in liquid pressure at the beginning of the pipeline.

In gravity flow systems, flow occurs due to elevation difference without any additional pump pressure. Thus, a pipeline from a storage tank on a hill to a delivery terminus below may not need any pump pressure at the tank.

However, the pipeline still needs to be designed to withstand pressure generated due to the static elevation difference.

The allowable operating pressure in a pipeline is defined as the maximum safe continuous pressure that the pipeline can be operated at. At this internal pressure the pipe material is stressed to some safe value below the yield strength of the pipe material. The stress in the pipe material consists of circumferential (or hoop) stress and longitudinal (or axial) stress. This is shown in Figure 4.1. It can be proven that the axial stress is one-half the value of the hoop stress. The hoop stress therefore controls the amount of internal pressure the pipeline can withstand. For pipelines transporting liquids, the hoop stress may be allowed to reach 72% of the pipe yield strength.

If pipe material has 60,000 psi yield strength, the safe internal operating pressure cannot exceed a value that results in a hoop stress of

$$0.72 \times 60,000 = 43,200 \text{ psi}$$

To ensure that the pipeline can be safely operated at a particular maximum allowable operating pressure (MAOP) we must test the pipeline using water, at a higher pressure.

The hydrostatic test pressure is a pressure higher than the allowable operating pressure. It is the pressure at which the pipeline is tested for a specified period of time, such as 4 hr (for aboveground piping) or 8 hr (for buried pipeline) as required by the pipeline design code or by the appropriate city or government regulations. In the United States, Department of Transportation (DOT) Code Part 195 applies. Generally, for liquid pipelines the hydrostatic test pressure is 25% higher than the MAOP. Thus, if the MAOP is 1000 psig, the pipeline will be hydrostatically tested at 1250 psig.

Calculation of internal design pressure in a pipeline is based on Barlow's equation for internal pressure in thin-walled cylindrical pipes, as discussed next.

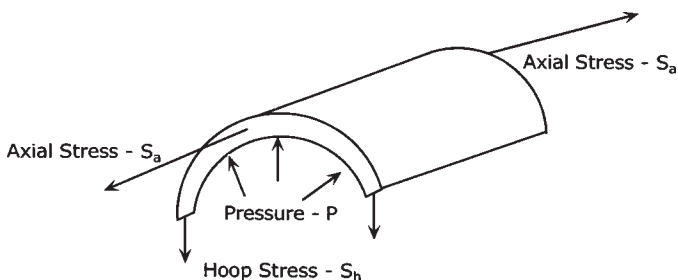


Figure 4.1 Hoop stress and axial stress in a pipe.

4.2 Barlow's Equation for Internal Pressure

The hoop stress or circumferential stress, S_h , in a thin-walled cylindrical pipe due to an internal pressure is calculated using the formula

$$S_h = PD/2 \quad (4.1)$$

where

S_h =Hoop stress, psi

P =Internal pressure, psi

D =Pipe outside diameter, in.

t =Pipe wall thickness, in.

Similarly, the axial (or longitudinal) stress, S_a , is

$$S_a = PD/4t \quad (4.2)$$

The above equations form the basis of Barlow's equation used to determine the allowable internal design pressure in a pipeline. As can be seen from Equations (4.1) and (4.2), the hoop stress is twice the longitudinal stress. The internal design pressure will therefore be based on the hoop stress (Equation 4.1).

Barlow's equation can be derived easily as follows: Consider one-half of a unit length of pipe as shown in [Figure 4.1](#). Due to internal pressure P , the bursting force on one-half the pipe is

$$P \times D \times 1$$

where the pressure P acts on a projected area $D \times 1$. This bursting force is exactly balanced by the hoop stress S_h acting along both edges of the pipe. Therefore,

$$S_h \times t \times 1 \times 2 = P \times D \times 1$$

Solving for S_h we get

$$S_h = PD/2t$$

Equation (4.2) for axial stress S_a is derived as follows. The axial stress S_a acts on an area of cross-section of pipe represented by πDt . This is balanced by the internal pressure P acting on the internal cross-sectional area of pipe $\pi D^2/4$. Equating the two we get

$$S_a \times \pi Dt = P \times \pi D^2/4$$

Solving for S_a , we get

$$S_a = PD/4t$$

In calculating the internal design pressure in liquid pipelines, we modify Barlow's equation slightly. The internal design pressure in a pipe is calculated in English units as follows:

$$P = \frac{2T \times S \times E \times F}{D} \quad (4.3)$$

where

P=Internal pipe design pressure, psig

D=Nominal pipe outside diameter, in.

T=Nominal pipe wall thickness, in.

S=Specified minimum yield strength (SMYS) of pipe material, psig

E=Seam joint factor, 1.0 for seamless and submerged arc welded (SAW) pipes (see [Table A.11](#) in Appendix A)

F=Design factor, usually 0.72 for liquid pipelines, except that a design factor of 0.60 is used for pipe, including risers, on a platform located off shore or on a platform in inland navigable waters, and 0.54 is used for pipe that has been subjected to cold expansion to meet the SMYS and subsequently heated, other than by welding or stress-relieving as a part of the welding, to a temperature higher than 900°F (482°C) for any period of time or to over 600°F (316°C) for more than 1 hr.

The above form of Barlow's equation may be found in Part 195 of DOT Code of Federal Regulations, Title 49 and ASME standard B31.4 for liquid pipelines. In SI units, Barlow's equation can be written as:

$$P = \frac{2T \times S \times E \times F}{D} \quad (4.4)$$

where

P=Pipe internal design pressure, kPa

D=Nominal pipe outside diameter, mm

T=Nominal pipe wall thickness, mm

S=Specified minimum yield strength (SMYS) of pipe material, kPa

E and F are defined under Equation (4.3)

In summary, Barlow's equation for internal pressure is based on calculation of the hoop stress (circumferential) in the pipe material. The hoop stress is the controlling stress within stressed pipe material, being twice the axial stress ([Figure 4.1](#)).

The strength of pipe material designated as specified minimum yield strength (SMYS) in Equations (4.3) and (4.4) depends on pipe material and grade. In the United States, steel pipeline material used in the oil and gas industry is manufactured in accordance with American Petroleum Institute (API) standards 5L and 5LX. For example, grades 5LX-42, 5LX-52, 5LX-60,

5LX-65, 5LX-70, and 5LX-80 are used commonly in pipeline applications. The numbers after 5LX above indicate the SMYS values in thousands of psi. Thus, 5LX-52 pipe has a minimum yield strength of 52,000 psi. The lowest grade of pipe material used is 5L Grade B, which has an SMYS of 35,000 psi. In addition, seamless steel pipe designated as ASTM A106 and Grade B pipe are also used for liquid pipeline systems. These have an SMYS value of 35,000 psi.

It is obvious from Barlow's equation (4.3) that, for a given pipe diameter, pipe material, and seam joint factor, the allowable internal pressure P is directly proportional to the pipe wall thickness. For example, a pipe of 16 in. diameter with a wall thickness of 0.250 in. made of 5LX-52 pipe has an allowable internal design pressure of 1170 psi calculated as follows:

$$P=(2 \times 0.250 \times 52,000 \times 1.0 \times 0.72)/16=1170 \text{ psig}$$

Therefore if the wall thickness is increased to 0.375 in. the allowable internal design pressure increases to

$$(0.375/0.250) \times 1170=1755 \text{ psig}$$

On the other hand, if the pipe material is changed to 5LX-70, keeping the wall thickness at 0.250 in., the new internal pressure is

$$(70,000/52,000)1170=1575 \text{ psig}$$

Note that we used Barlow's equation to calculate the allowable internal pressure based upon the pipe material being stressed to 72% of SMYS. In some situations more stringent city or government regulations may require that the pipe be operated at a lower pressure. Thus, instead of using a 72% factor in Equation (4.3) we may be required to use a more conservative factor (lower number) in place of $F=0.72$. As an example, in certain areas of Los Angeles, liquid pipelines are only allowed to operate at a 66% factor instead of the 72% factor. Therefore, in the earlier example, the 16 in./0.250 in./X52 pipeline can only be operated at

$$1170(66/72)=1073 \text{ psig}$$

As mentioned before, in order to operate a pipeline at 1170 psig, it must be hydrostatically tested at 25% higher pressure. Since 1170 psig internal pressure is based on the pipe material being stressed to 72% of SMYS, the hydrostatic test pressure will cause the hoop stress to reach

$$1.25(72)=90\% \text{ of SMYS}$$

Generally, the hydrostatic test pressure is specified as a range of pressures, such as 90% SMYS to 95% SMYS. This is called the hydrotest

pressure envelope. Therefore, in the present example the hydrotest pressure range is

$$\begin{aligned} 1.25(1170) &= 1463 \text{ psig} && \text{lower limit (90\% SMYS)} \\ (95/90)1463 &= 1544 \text{ psig} && \text{higher limit (95\% SMYS)} \end{aligned}$$

To summarize, a pipeline with an MAOP of 1170 psig needs to be hydrotested at a pressure range of 1463 psig to 1544 psig. According to the design code, the test pressure will be held for a minimum 4 hr for aboveground pipelines and 8 hr for buried pipelines.

In calculating the allowable internal pressure in older pipelines, consideration must be given to wall thickness reduction due to corrosion over the life of the pipeline. A pipeline that was installed 25 years ago with 0.250 in. wall thickness may have reduced in wall thickness to 0.200 in. or less due to corrosion. Therefore, the allowable internal pressure will have to be reduced in the ratio of the wall thickness, compared with the original design pressure.

4.3 Line Fill Volume and Batches

Frequently we need to know how much liquid is contained in a pipeline between two points along its length, such as between valves or pump stations.

For a circular pipe, we can calculate the volume of a given length of pipe by multiplying the internal cross-sectional area by the pipe length. If the pipe internal diameter is D in. and the length is L ft, the volume of this length of pipe is

$$V = 0.7854(D^2/144)L \quad (4.5)$$

where

$$V = \text{Volume, ft}^3$$

Simplifying,

$$V = 5.4542 \times 10^{-3} D^2 L \quad (4.6)$$

We will now restate this equation in terms of conventional pipeline units, such as the volume in bbl in a mile of pipe. The quantity of liquid contained in a mile of pipe, also called the line fill volume, is calculated as follows:

$$V_L = 5.129(D)^2 \quad (4.7)$$

where

V_L = Line fill volume of pipe, bbl/mile

D = Pipe internal diameter, in.

In SI units we can express the line fill volume per km of pipe as follows:

$$V_L = 7.855 \times 10^{-4} D^2 \tag{4.8}$$

where

V_L = Line fill volume, m³/km

D = Pipe internal diameter, mm

Using Equation (4.7), a pipeline 100 miles long, 16 in. diameter and 0.250 in. wall thickness, has a line fill volume of

$$5.129(15.5)^2 (100) = 123,224 \text{ bbl}$$

Many crude oil and refined product pipelines operate in a batched mode, in which multiple products are simultaneously pumped through the pipeline as batches. For example, 50,000 bbl of product C will enter the pipeline followed by 30,000 bbl of product B and 40,000 bbl of product A. If the line fill volume of the pipeline is 120,000 bbl, an instantaneous snapshot condition of a batched pipeline is as shown in Figure 4.2.

Example Problem 4.1

A 50 mile pipeline consists of 20 miles of pipe of 16 in. diameter and 0.375 in. wall thickness followed by 30 miles of pipe of 14 in. diameter and 0.250 in. wall thickness. Calculate the total volume contained in the 50 miles of pipeline.

Solution

Using Equation (4.7) we get, for the 16 in. pipeline

$$\text{Volume per mile} = 5.129(15.25)^2 = 1192.81 \text{ bbl/mile}$$

And for the 14 in. pipeline

$$\text{Volume per mile} = 5.129(13.5)^2 = 934.76 \text{ bbl/mile}$$

Total line fill volume is

$$20 \times 1192.81 + 30 \times 934.76 = 51,899 \text{ bbl}$$

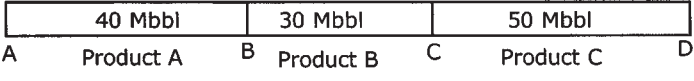


Figure 4.2 Batched pipeline.

Example Problem 4.2

A pipeline 100 km long is 500 mm outside diameter and 12 mm wall thickness. If batches of three liquids A (3000m^3), B (5000m^3) and C occupy the pipe, at a particular instant, calculate the interface locations of the batches, considering the origin of the pipeline to be at 0.0 km.

Solution

Using Equation (4.8) we get the line fill volume per km to be

$$V_L = 7.855 \times 10^{-4} (500 - 24)^2 = 177.9754 \text{ m}^3/\text{km}$$

The first batch A will start at 0.0 km and will end at a distance of

$$3000/177.9754 = 16.86 \text{ km}$$

The second batch B starts at 16.86 km and ends at

$$16.86 + (5000/177.9754) = 44.95 \text{ km}$$

The third batch C starts at 44.95 km and ends at 100 km.

The total volume in the pipe is

$$177.9754 \times 100 = 17,798 \text{ m}^3$$

Thus the volume of the third batch C is

$$17,798 - 3000 - 5000 = 9798 \text{ m}^3$$

It can thus be seen that line fill volume calculation is important when dealing with batched pipelines. We need to know the boundaries of each liquid batch, so that the correct liquid properties can be used to calculate pressure drops for each batch.

The total pressure drop in a batched pipeline would be calculated by adding up the individual pressure drops for each batch. Since intermixing of the batches is not desirable, batched pipelines must run in turbulent flow. In laminar flow there will be extensive mixing of the batches, which defeats the purpose of keeping each product separate; so that at the end of the pipeline each product may be diverted into a separate tank. Some intermixing will occur at the product interfaces and this contaminated liquid is generally pumped into a slop tank at the end of the pipeline and may be blended with a less critical product. The amount of contamination that occurs at the batch interface depends on the physical properties of the batched products, batch length, and Reynolds number.

4.4 Summary

In this chapter we discussed how allowable internal pressure in a pipeline is calculated depending on pipe size and material. We showed that for pipe under internal pressure the hoop stress in the pipe material will be the controlling factor. The importance of design factor in selecting pipe wall thickness was illustrated using an example. Based on Barlow's equation, the internal design pressure calculation as recommended by ASME standard B31.4 and US Code of Federal Regulation, Part 195 of the DOT was illustrated. The need for hydrostatic testing pipelines for safe operation was discussed. The line fill volume calculation was introduced and its importance in batched pipelines was shown using an example.

4.5 Problems

- 4.5.1 Calculate the allowable internal design pressure at 72% design factor for an 18 in. pipeline, wall thickness 0.375 in. and pipe material API 5LX-46. What is the hydrotest pressure range for this pipeline?
- 4.5.2 It has been determined that the design pressure for a storage tank piping system is 720 psi. If API 5L grade B pipe is used, what minimum wall thickness is required for 14 in. pipe?
- 4.5.3 In Problem 4.5.2, if the pressure rating were increased to ANSI 600 (1440 psi), calculate the pipe wall thickness required with 14 in. pipe if high-strength 5LX-52 pipe is used. What is the minimum hydrotest pressure for this system?
- 4.5.4 Determine the volume of liquid contained in a mile of 14 in. pipeline with a wall thickness of 0.281 in.